Mathematics-III (Math 2123) Beginner-Friendly Study Guide

Topics + Basics + Problem-Solving Procedures + Complete Solutions

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Based on the uploaded examination paper:

- Khulna University of Engineering & Technology (KUET)
- Department of Building Engineering and Construction Management
- B.Sc. Engineering 2nd Year 1st Term Regular Examination, 2025
- Mathematics-III (Math 2123)

Designed for a beginner. This guide does not assume you are already comfortable with the topics. Each of the 8 question sets starts with the underlying topic, explains the minimum theory you need, gives a repeatable procedure, and then solves every visible sub-question step by step.

Note on orientation: In Question 8(a), the normal is taken in the positive direction of $(2, 3, 6)$; the opposite normal changes the sign. In Question 8(c), counter-clockwise orientation is taken as positive; clockwise changes the sign.

Contents

1. How to Use This Guide

BASIC IDEA

The paper is not eight unrelated questions. It is mainly four connected blocks: matrices/linear algebra (Sets 1-4), Laplace transforms (Sets 5-6), vector differential calculus (Sets 6-7), and vector integral calculus (Set 8). Learn the procedures, not only the final answers.

Set	Main topic	What you should be able to do
1	Matrix fundamentals	Identify special matrices, decompose a matrix, solve $AX = B$
2	Inverse, consistency, dependence	Find A^{-1}, classify linear systems, test vectors
3	Rank and eigenvalues	Row-reduce, find rank, eigenvalues and eigenvectors
4	Vectors + Cayley-Hamilton	Velocity/acceleration, dot-product work, matrix polynomial
5	Laplace transform	Shift rules, periodic functions, unit step, inverse LT
6	Convolution + ODE + gradient	Convolution integral, solve IVP, compute gradient
7	Divergence, curl, potential	Expansion/compression, irrotational/conservative fields
8	Surface/volume/line integrals	Flux, vector triple integral, Green's theorem

EXAM TIP

When practicing, cover the worked solution. Read only the "procedure" box, then try the problem. If you get stuck, uncover one step at a time. This trains exam-solving rather than memorization.

2. Set 1 - Matrix Fundamentals and Solving Linear Equations

Topic map

- Special matrices: lower triangular, symmetric, nilpotent, idempotent
- Testing idempotency by calculating A^2
- Decomposition into symmetric and skew-symmetric parts
- Solving a linear system in matrix form $AX = B$

BASIC IDEA

A matrix is a rectangular arrangement of numbers. Most questions here depend on one simple operation: transpose. The transpose A^T is obtained by turning rows into columns.

1(a) Definitions with examples

(i) Lower triangular matrix

A square matrix is **lower triangular** if every entry above the main diagonal is zero:

$$A = \begin{bmatrix} 2 & 0 & 0 \\ -1 & 3 & 0 \\ 4 & 5 & 1 \end{bmatrix}.$$

The entries below the diagonal may be zero or nonzero.

(ii) Symmetric matrix

A square matrix is **symmetric** if

$$A^T = A.$$

Example:

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 4 \end{bmatrix}.$$

The entry in position (i, j) equals the entry in (j, i).

(iii) Nilpotent matrix

A square matrix is **nilpotent** if for some positive integer k,

$$A^k = 0.$$

Example:

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \qquad A^2 = 0.$$

(iv) Idempotent matrix

A square matrix is **idempotent** if

$$A^2 = A.$$

Example:

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \qquad A^2 = A.$$

COMMON MISTAKE

Nilpotent means a power of the matrix becomes the zero matrix. Idempotent means squaring the matrix gives the same matrix. Do not confuse these two definitions.

1(b) Test whether the given matrix is idempotent

$$A = \begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 5 \end{bmatrix}.$$

PROBLEM-SOLVING PROCEDURE

For an idempotent test, calculate $A^2 = A \cdot A$. If every element of A^2 matches A, it is idempotent. If even one element differs, it is not.

Multiplying,

$$A^2 = \begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 5 \end{bmatrix} \begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 5 \end{bmatrix} = \begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 5 \end{bmatrix} = A.$$

Therefore the matrix is idempotent.

Final answer: $A^2 = A$, so A is idempotent.

1(c) Express A as symmetric + skew-symmetric

$$A = \begin{bmatrix} 3 & 1 & 0 \\ 5 & 0 & 3 \\ 2 & 4 & 7 \end{bmatrix}.$$

BASIC IDEA

Every square matrix can be written uniquely as $A = S + K$, where $S^T = S$ (symmetric) and $K^T = -K$ (skew-symmetric).

Use the standard formulas

$$S = \frac{1}{2}(A + A^T), \qquad K = \frac{1}{2}(A - A^T).$$

First,

$$A^T = \begin{bmatrix} 3 & 5 & 2 \\ 1 & 0 & 4 \\ 0 & 3 & 7 \end{bmatrix}.$$

Hence

$$S = \frac{1}{2}\begin{bmatrix} 6 & 6 & 2 \\ 6 & 0 & 7 \\ 2 & 7 & 14 \end{bmatrix} = \begin{bmatrix} 3 & 3 & 1 \\ 3 & 0 & \frac{7}{2} \\ 1 & \frac{7}{2} & 7 \end{bmatrix},$$

and

$$K = \frac{1}{2}\begin{bmatrix} 0 & -4 & -2 \\ 4 & 0 & -1 \\ 2 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -2 & -1 \\ 2 & 0 & -\frac{1}{2} \\ 1 & \frac{1}{2} & 0 \end{bmatrix}.$$

Therefore

$$\boxed{A = S + K}.$$

1(d) Solve the system by matrix method

$$\begin{aligned} 4x - 5y + z &= 2, \\ 3x + y - 2z &= 9, \\ x + 4y + z &= 5. \end{aligned}$$

PROBLEM-SOLVING PROCEDURE

Write the equations as $AX = B$. If $\det A \neq 0$, then A^{-1} exists and $X = A^{-1}B$. In an exam, showing A, X, B, the inverse, and the multiplication is a clean "matrix method" solution.

$$A = \begin{bmatrix} 4 & -5 & 1 \\ 3 & 1 & -2 \\ 1 & 4 & 1 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} 2 \\ 9 \\ 5 \end{bmatrix}.$$

Here

$$\det A = 72 \neq 0,$$

so the inverse exists. A convenient inverse is

$$A^{-1} = \begin{bmatrix} \frac{1}{8} & \frac{1}{8} & \frac{1}{8} \\ -\frac{5}{72} & \frac{1}{24} & \frac{11}{72} \\ \frac{11}{72} & -\frac{7}{24} & \frac{19}{72} \end{bmatrix}.$$

Thus

$$X = A^{-1}B = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}.$$

Therefore

$$\boxed{x = 2, \quad y = 1, \quad z = -1}.$$

EXAM TIP

Always substitute the final values into at least one original equation. Here $4(2) - 5(1) - 1 = 2$, so the answer is consistent.

3. Set 2 - Matrix Inverse, Consistency of Systems, Linear Dependence

Topic map

- When a matrix has an inverse
- Finding the inverse using elementary row operations
- Classifying a system: unique / no solution / infinitely many solutions
- Testing whether vectors are linearly dependent

2(a) Does an inverse exist for all matrices? Find the inverse if possible

$$A = \begin{bmatrix} 1 & 20 & 2 \\ 2 & -1 & 3 \\ 4 & 1 & 8 \end{bmatrix}.$$

BASIC IDEA

A two-sided inverse exists only for a square nonsingular matrix. For a square matrix, A^{-1} exists exactly when $\det A \neq 0$.

For this matrix,

$$\det A = -79 \neq 0,$$

so A is invertible.

PROBLEM-SOLVING PROCEDURE

To find A^{-1} by elementary transformation, write the augmented matrix $[A \mid I]$ and use row operations until the left side becomes I. The right side then becomes A^{-1}.

Start with

$$\left[\begin{array}{ccc|ccc} 1 & 20 & 2 & 1 & 0 & 0 \\ 2 & -1 & 3 & 0 & 1 & 0 \\ 4 & 1 & 8 & 0 & 0 & 1 \end{array}\right].$$

Apply

$$R_2 \leftarrow R_2 - 2R_1, \qquad R_3 \leftarrow R_3 - 4R_1,$$

which gives

$$\left[\begin{array}{ccc|ccc} 1 & 20 & 2 & 1 & 0 & 0 \\ 0 & -41 & -1 & -2 & 1 & 0 \\ 0 & -79 & 0 & -4 & 0 & 1 \end{array}\right].$$

Now

$$R_3 \leftarrow -\frac{1}{79}R_3$$

gives

$$R_3 = \left[0, 1, 0 \mid \frac{4}{79}, 0, -\frac{1}{79}\right].$$

Then

$$R_2 \leftarrow R_2 + 41R_3, \qquad R_2 \leftarrow -R_2,$$

so

$$R_2 = \left[0, 0, 1 \mid -\frac{6}{79}, -1, \frac{41}{79}\right].$$

Eliminate the two nonzero entries in the first row:

$$R_1 \leftarrow R_1 - 2R_2, \qquad R_1 \leftarrow R_1 - 20R_3.$$

Finally swap R_2 and R_3 to put the identity matrix in standard order. Therefore

$$\boxed{A^{-1} = \begin{bmatrix} \frac{11}{79} & 2 & -\frac{62}{79} \\ \frac{4}{79} & 0 & -\frac{1}{79} \\ -\frac{6}{79} & -1 & \frac{41}{79} \end{bmatrix}}$$

2(b) Values of a, b for unique / no / infinite solutions

$$\begin{aligned} 3x - 2y + z &= b, \\ 5x - 8y + 9z &= 3, \\ 2x + y + az &= -1. \end{aligned}$$

The coefficient matrix is

$$A = \begin{bmatrix} 3 & -2 & 1 \\ 5 & -8 & 9 \\ 2 & 1 & a \end{bmatrix}.$$

Its determinant is

$$\det A = -14(a+3).$$

PROBLEM-SOLVING PROCEDURE

First test the determinant. If $\det A \neq 0$, there is exactly one solution. If $\det A = 0$, the system is singular, so then test consistency by comparing the row relations of the coefficient matrix with the right-hand side.

Case 1: $a \neq -3$. Then $\det A \neq 0$. Hence a unique solution exists for every b.

Case 2: $a = -3$. The coefficient rows satisfy

$$-3R_1 + R_2 + 2R_3 = 0.$$

For consistency, the same combination of the right-hand sides must also be zero:

$$-3b + 3 + 2(-1) = 0.$$

Thus

$$-3b + 1 = 0 \quad \Rightarrow \quad b = \frac{1}{3}.$$

At $a = -3$, the coefficient rank is $2 < 3$.

Therefore:

Unique solution:	$a \neq -3 \quad (\text{any } b)$,
No solution:	$a = -3,\ b \neq \frac{1}{3}$,
Infinitely many solutions:	$a = -3,\ b = \frac{1}{3}$.

COMMON MISTAKE

When the determinant is zero, do not immediately say "no solution." A singular system may have no solution or infinitely many solutions. You must check consistency.

2(c) Linear dependence and independence

$$X_1 = (1, 2, 4), \quad X_2 = (2, -1, 3), \quad X_3 = (0, 1, 2).$$

BASIC IDEA

> Vectors X_1, X_2, X_3 are linearly dependent if there exist constants c_1, c_2, c_3, not all zero, such that $c_1X_1 + c_2X_2 + c_3X_3 = 0$. They are independent if the only solution is $c_1 = c_2 = c_3 = 0$.

Put the vectors as columns:

$$M = \begin{bmatrix} 1 & 2 & 0 \\ 2 & -1 & 1 \\ 4 & 3 & 2 \end{bmatrix}.$$

Then

$$\det M = -5 \neq 0.$$

Therefore the only solution of $M\mathbf{c} = 0$ is the trivial solution.

Final answer: X_1, X_2, X_3 are linearly independent.

4. Set 3 - Rank, Echelon Form, Eigenvalues and Eigenvectors

3(a) Rank of a matrix

$$A = \begin{bmatrix} 2 & -6 & -2 & -3 \\ 5 & -13 & -4 & -7 \\ -1 & 4 & 1 & 2 \\ 0 & 1 & 0 & 1 \end{bmatrix}.$$

BASIC IDEA

The rank is the number of independent rows (equivalently independent columns). In row-echelon form, the rank equals the number of nonzero rows, or the number of pivots.

PROBLEM-SOLVING PROCEDURE

Use elementary row operations to create a staircase of pivots. Count the pivots for the rank. Continue upward elimination to get reduced row-echelon (canonical) form. For a full-rank square matrix, the reduced form and the normal form are both the identity matrix.

Start by interchanging R_1 and R_3, then multiply the new R_1 by -1:

$$R_1 = [1, -4, -1, -2].$$

Now use

$$R_2 \leftarrow R_2 - 5R_1, \qquad R_3 \leftarrow R_3 - 2R_1.$$

This gives rows

$$R_2 = [0, 7, 1, 3], \qquad R_3 = [0, 2, 0, 1], \qquad R_4 = [0, 1, 0, 1].$$

Swap R_2 and R_4, then perform

$$R_3 \leftarrow R_3 - 2R_2, \qquad R_4 \leftarrow R_4 - 7R_2,$$

and swap the last two rows. One convenient echelon form is

$$\boxed{\begin{bmatrix} 1 & -4 & -1 & -2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -4 \\ 0 & 0 & 0 & -1 \end{bmatrix}}$$

There are four pivots, so already

$$\operatorname{rank}(A) = 4.$$

To continue to reduced row-echelon form, take

$$R_4 \leftarrow -R_4, \quad R_3 \leftarrow R_3 + 4R_4, \quad R_2 \leftarrow R_2 - R_4,$$

then eliminate the remaining entries in R_1. The result is

$$\boxed{\mathrm{RREF}(A) = I_4}.$$

Thus the canonical form is I_4. Since this matrix has full rank, its normal form is also

$$\boxed{I_4}.$$

Final answer: $\operatorname{rank}(A) = 4$.

EXAM TIP

Some textbooks use the words “canonical form” and “normal form” slightly differently. In this question the matrix has full rank, so both reductions end at I_4; the final rank is unambiguous.

3(b) Eigenvalues and eigenvectors

$$A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}.$$

BASIC IDEA

An eigenvector is a nonzero vector v whose direction is unchanged by the matrix: $Av = \lambda v$. The scalar λ is the eigenvalue. Rearranging gives $(A - \lambda I)v = 0$, which has a nonzero solution only when $\det(A - \lambda I) = 0$.

PROBLEM-SOLVING PROCEDURE

Step 1: form the characteristic equation $\det(A - \lambda I) = 0$. Step 2: factor it to get eigenvalues. Step 3: for each eigenvalue, solve $(A - \lambda I)v = 0$.

The characteristic polynomial can be written as

$$\det(\lambda I - A) = (\lambda - 8)(\lambda - 2)^2.$$

Therefore

$$\boxed{\lambda = 8, \quad \lambda = 2 \text{ (double)}}.$$

For $\lambda = 8$,

$$(A - 8I)v = 0$$

gives, for example,

$$\boxed{v = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}}.$$

Indeed,

$$A \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} = 8 \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}.$$

For $\lambda = 2$, the equations reduce to

$$2x - y + z = 0 \quad \Rightarrow \quad y = 2x + z.$$

Two independent eigenvectors are

$$\boxed{\begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \qquad \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}}.$$

5. Set 4 - Vector Motion, Work, and Cayley-Hamilton Theorem

4(a) Velocity and acceleration of a particle

The position is

$$\mathbf{r}(t) = (e^{-t},\ 2\cos 3t,\ 2\sin 3t).$$

BASIC IDEA

Velocity is the first derivative of position: $\mathbf{v} = d\mathbf{r}/dt$. Acceleration is the derivative of velocity: $\mathbf{a} = d\mathbf{v}/dt = d^2\mathbf{r}/dt^2$. The magnitude of a vector (p, q, r) is $\sqrt{p^2 + q^2 + r^2}$.

Differentiate:

$$\mathbf{v}(t) = \left(-e^{-t},\ -6\sin 3t,\ 6\cos 3t\right).$$

At $t = 0$,

$$\mathbf{v}(0) = (-1, 0, 6),$$

so

$$|\mathbf{v}(0)| = \sqrt{(-1)^2 + 0^2 + 6^2} = \boxed{\sqrt{37}}.$$

Differentiate again:

$$\mathbf{a}(t) = \left(e^{-t},\ -18\cos 3t,\ -18\sin 3t\right).$$

At $t = 0$,

$$\mathbf{a}(0) = (1, -18, 0),$$

so

$$|\mathbf{a}(0)| = \sqrt{1 + 324} = \sqrt{325} = \boxed{5\sqrt{13}}.$$

4(b) Work done by a constant force

Displacement:

$$\mathbf{r} = 3\hat{\imath} + \hat{\jmath} - 5\hat{k},$$

force:

$$\mathbf{F} = 2\hat{\imath} - \hat{\jmath} - \hat{k}.$$

BASIC IDEA

For a constant force, work is the dot product: $W = \mathbf{F} \cdot \mathbf{r}$. Multiply matching components and add.

$$W = (2)(3) + (-1)(1) + (-1)(-5) = 6 - 1 + 5 = \boxed{10}.$$

4(c) Cayley-Hamilton theorem and A^{-1}

$$A = \begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix}.$$

BASIC IDEA

Cayley-Hamilton theorem says that every square matrix satisfies its own characteristic equation. If the characteristic polynomial is $p(\lambda)$, then $p(A) = 0$. This often lets us express A^{-1} using lower powers of A.

The characteristic equation is

$$\det(\lambda I - A) = \begin{vmatrix} \lambda - 1 & -2 \\ 1 & \lambda - 1 \end{vmatrix} = (\lambda - 1)^2 + 2 = \lambda^2 - 2\lambda + 3 = 0.$$

By Cayley-Hamilton,

$$A^2 - 2A + 3I = 0.$$

Now

$$A^2 = \begin{bmatrix} -1 & 4 \\ -2 & -1 \end{bmatrix},$$

and substituting confirms that $A^2 - 2A + 3I$ is the zero matrix.

To find the inverse, multiply the Cayley-Hamilton equation by A^{-1}:

$$A - 2I + 3A^{-1} = 0.$$

Hence

$$A^{-1} = \frac{2I - A}{3} = \boxed{\begin{bmatrix} \frac{1}{3} & -\frac{2}{3} \\ \frac{1}{3} & \frac{1}{3} \end{bmatrix}}.$$

6. Set 5 - Laplace Transform, Periodic Functions, Unit Step, Inverse Transform

Essential Laplace formulas

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t)\,dt.$$

Useful formulas:

$$\mathcal{L}\{\sin at\} = \frac{a}{s^2+a^2}, \quad \mathcal{L}\{tf(t)\} = -\frac{d}{ds}F(s), \quad \mathcal{L}\{e^{-at}f(t)\} = F(s+a).$$

5(a) Sufficient conditions for existence of a Laplace transform

A standard sufficient set of conditions is:

1. $f(t)$ is piecewise continuous on every finite interval $0 \le t \le T$.
2. $f(t)$ is of exponential order: there exist constants M, c, T such that $|f(t)| \le Me^{ct}$ for $t \ge T$.

Under these conditions, the Laplace transform exists for sufficiently large s.

5(a)(i) $F(t) = (t+2)e^{-3t}\sin 5t$

First ignore e^{-3t} and transform $(t+2)\sin 5t$.

$$\mathcal{L}\{\sin 5t\} = \frac{5}{s^2+25}.$$

Using $\mathcal{L}\{tf(t)\} = -F'(s)$,

$$\mathcal{L}\{t\sin 5t\} = -\frac{d}{ds}\left(\frac{5}{s^2+25}\right) = \frac{10s}{(s^2+25)^2}.$$

Therefore

$$\mathcal{L}\{(t+2)\sin 5t\} = \frac{10s}{(s^2+25)^2} + \frac{10}{s^2+25}.$$

Now use the first shifting rule $s \mapsto s+3$:

$$\boxed{\mathcal{L}\{(t+2)e^{-3t}\sin 5t\} = \frac{10(s+3)}{((s+3)^2+25)^2} + \frac{10}{(s+3)^2+25}}$$

5(a)(ii) Periodic function

$$G(t) = \begin{cases} t, & 0<t<2, \\ 0, & 2<t<4, \end{cases} \qquad G(t+4) = G(t).$$

BASIC IDEA

For a periodic function of period T, $\mathcal{L}\{f\} = \dfrac{\int_0^T e^{-st}f(t)dt}{1-e^{-sT}}$. You only integrate over one period.

Here $T = 4$:

$$\mathcal{L}\{G\} = \frac{\int_0^4 e^{-st}G(t)dt}{1-e^{-4s}} = \frac{\int_0^2 te^{-st}dt}{1-e^{-4s}}.$$

Now

$$\int_0^2 te^{-st}dt = \frac{1-(1+2s)e^{-2s}}{s^2}.$$

Hence

$$\boxed{\mathcal{L}\{G(t)\} = \frac{1-(1+2s)e^{-2s}}{s^2(1-e^{-4s})}}$$

5(b) Unit step function and beam loading

$$p(t) = \begin{cases} 5t, & 0 < t < 4, \\ 20, & t \geq 4. \end{cases}$$

BASIC IDEA

The unit step $u(t-a)$ equals 0 before $t = a$ and 1 from $t = a$ onward. It is used to "switch on" a correction at a specified time.

Before $t = 4$, the formula $5t$ is correct. After $t = 4$, we need to replace $5t$ by 20. The required correction is

$$20 - 5t = -5(t-4).$$

Thus

$$\boxed{p(t) = 5t - 5(t-4)u(t-4)}.$$

Using

$$\mathcal{L}\{(t-a)u(t-a)\} = e^{-as}\frac{1}{s^2},$$

we get

$$\boxed{P(s) = \frac{5}{s^2} - \frac{5e^{-4s}}{s^2} = \frac{5(1-e^{-4s})}{s^2}}.$$

5(c) Inverse Laplace transforms

5(c)(i) $f(s) = \dfrac{e^{-5s}}{(s-2)^4}$

First,

$$\mathcal{L}^{-1}\left\{\frac{1}{(s-2)^4}\right\} = \frac{t^3}{3!}e^{2t} = \frac{t^3e^{2t}}{6}.$$

The factor e^{-5s} produces a delay of 5:

$$\boxed{\mathcal{L}^{-1}\left\{\frac{e^{-5s}}{(s-2)^4}\right\} = \frac{(t-5)^3}{6}e^{2(t-5)}u(t-5)}.$$

5(c)(ii) $g(s) = \dfrac{2s^2-4}{(s+1)(s-2)(s-3)}$

Use partial fractions:

$$\frac{2s^2-4}{(s+1)(s-2)(s-3)} = -\frac{1}{6(s+1)} - \frac{4}{3(s-2)} + \frac{7}{2(s-3)}.$$

Therefore

$$\boxed{g(t) = -\frac{1}{6}e^{-t} - \frac{4}{3}e^{2t} + \frac{7}{2}e^{3t}}.$$

COMMON MISTAKE

For inverse transforms, expressions like $1/(s-a)$ produce e^{at}, not e^{-at}. The sign comes from the denominator: $\mathcal{L}\{e^{at}\} = 1/(s-a)$.

7. Set 6 - Convolution, Laplace Solution of an ODE, and Gradient

6(a) Convolution theorem

BASIC IDEA

If $F(s) = \mathcal{L}\{f(t)\}$ and $G(s) = \mathcal{L}\{g(t)\}$, then $\mathcal{L}^{-1}\{F(s)G(s)\} = (f*g)(t)$, where $(f*g)(t) = \int_0^t f(\tau)g(t-\tau)d\tau$.

Evaluate

$$\mathcal{L}^{-1}\left\{\frac{1}{(s+2)^2(s-2)}\right\}.$$

Choose

$$F(s) = \frac{1}{(s+2)^2} \quad \Rightarrow \quad f(t) = te^{-2t},$$

$$G(s) = \frac{1}{s-2} \quad \Rightarrow \quad g(t) = e^{2t}.$$

Then

$$(f*g)(t) = \int_0^t \tau e^{-2\tau}e^{2(t-\tau)}d\tau = e^{2t}\int_0^t \tau e^{-4\tau}d\tau.$$

Since

$$\int_0^t \tau e^{-4\tau}d\tau = \frac{1}{16} - \left(\frac{t}{4} + \frac{1}{16}\right)e^{-4t},$$

we obtain

$$\boxed{\mathcal{L}^{-1}\left\{\frac{1}{(s+2)^2(s-2)}\right\} = \frac{e^{2t} - e^{-2t}}{16} - \frac{t}{4}e^{-2t}}$$

6(b) Earthquake displacement using Laplace transform

$$x'' + 2x' + x = e^{-t}, \qquad x(0) = 0, \quad x'(0) = 1.$$

PROBLEM-SOLVING PROCEDURE

Take the Laplace transform of every term. Use $\mathcal{L}\{x'\} = sX - x(0)$ and $\mathcal{L}\{x''\} = s^2X - sx(0) - x'(0)$. Solve the resulting algebraic equation for $X(s)$, then invert.

Taking transforms,

$$(s^2X - 1) + 2(sX) + X = \frac{1}{s+1}.$$

Thus

$$(s+1)^2X - 1 = \frac{1}{s+1},$$

so

$$X(s) = \frac{s+2}{(s+1)^3}.$$

Write $s+2 = (s+1)+1$:

$$X(s) = \frac{1}{(s+1)^2} + \frac{1}{(s+1)^3}.$$

Hence

$$\boxed{x(t) = te^{-t} + \frac{t^2}{2}e^{-t} = e^{-t}\left(t + \frac{t^2}{2}\right)}.$$

6(c) Temperature gradient

$$T(x, y, z) = x^2ye^{-z} + 3xy^2 + z^3.$$

BASIC IDEA

The gradient $\nabla T = (T_x, T_y, T_z)$ points in the direction of greatest increase of T. Its magnitude $|\nabla T|$ is the maximum rate of increase.

Compute the partial derivatives:

$$T_x = 2xye^{-z} + 3y^2,$$
$$T_y = x^2e^{-z} + 6xy,$$
$$T_z = -x^2ye^{-z} + 3z^2.$$

At $(1, 2, 0)$,

$$\nabla T = (16, 13, -2).$$

Therefore the magnitude is

$$\boxed{|\nabla T| = \sqrt{16^2 + 13^2 + (-2)^2} = \sqrt{429}}.$$

The direction of maximum heat increase is the gradient direction. A unit vector is

$$\boxed{\hat{u} = \frac{16\hat{i} + 13\hat{j} - 2\hat{k}}{\sqrt{429}}}.$$

8. Set 7 - Divergence, Curl, Irrotational and Conservative Fields

7(a) Expansion or compression of airflow

$$\mathbf{V} = (x^2y + yz)\hat{\imath} + (xy^2 + xz)\hat{\jmath} + (yz^2 + x^2z)\hat{k}.$$

BASIC IDEA

Divergence measures net "spreading out" of a vector field. If $\nabla \cdot \mathbf{V} > 0$, the field has local expansion; if it is negative, local compression; if zero, neither.

$$\nabla \cdot \mathbf{V} = \frac{\partial}{\partial x}(x^2y + yz) + \frac{\partial}{\partial y}(xy^2 + xz) + \frac{\partial}{\partial z}(yz^2 + x^2z).$$

Thus

$$\nabla \cdot \mathbf{V} = 2xy + 2xy + (2yz + x^2).$$

At $(1, 2, 1)$,

$$\nabla \cdot \mathbf{V} = 4 + 4 + 4 + 1 = \boxed{13}.$$

Because $13 > 0$, the airflow has **expansion** at the point.

7(b) Test whether the velocity field is irrotational and find potential

$$\mathbf{V} = (x^3 + yz)\hat{\imath} + (y^3 + xz)\hat{\jmath} + (z^3 + xy)\hat{k}.$$

BASIC IDEA

A field is irrotational if $\nabla \times \mathbf{V} = 0$. In a simply connected region, an irrotational field has a scalar potential ϕ such that $\mathbf{V} = \nabla\phi$.

Let

$$P = x^3 + yz, \quad Q = y^3 + xz, \quad R = z^3 + xy.$$

The curl is

$$\nabla \times \mathbf{V} = (R_y - Q_z,\ P_z - R_x,\ Q_x - P_y).$$

Here

$$R_y = x = Q_z, \qquad P_z = y = R_x, \qquad Q_x = z = P_y.$$

Therefore

$$\boxed{\nabla \times \mathbf{V} = 0},$$

so the field is irrotational.

To find the potential, integrate P with respect to x:

$$\phi = \frac{x^4}{4} + xyz + g(y, z).$$

Differentiate with respect to y and compare with Q:

$$\phi_y = xz + g_y = y^3 + xz \quad \Rightarrow \quad g_y = y^3.$$

Hence

$$g = \frac{y^4}{4} + h(z).$$

Now compare ϕ_z with R:

$$\phi_z = xy + h'(z) = z^3 + xy \quad \Rightarrow \quad h'(z) = z^3.$$

Thus

$$\boxed{\phi(x, y, z) = \frac{x^4 + y^4 + z^4}{4} + xyz + C}.$$

7(c) Conservative force field and work

$$\mathbf{F} = (2xy + z^2)\hat{\imath} + (x^2 + 2yz)\hat{\jmath} + (2xz + y^2)\hat{k}.$$

Check the cross-partials:

$$P_y = 2x = Q_x, \qquad P_z = 2z = R_x, \qquad Q_z = 2y = R_y.$$

Hence

$$\boxed{\nabla \times \mathbf{F} = 0},$$

so the field is conservative.

Find a potential. Integrating $P = 2xy + z^2$ with respect to x,

$$\phi = x^2y + xz^2 + g(y, z).$$

Using $\phi_y = Q$ gives

$$g_y = 2yz \quad \Rightarrow \quad g = y^2z + h(z).$$

Using $\phi_z = R$ gives $h'(z) = 0$. Thus

$$\phi = x^2y + xz^2 + y^2z + C.$$

For a conservative field, work depends only on the endpoints:

$$W = \phi(2, 1, 1) - \phi(1, 0, 0).$$

Now

$$\phi(2, 1, 1) = 4 + 2 + 1 = 7, \qquad \phi(1, 0, 0) = 0.$$

Therefore

$$\boxed{W = 7}.$$

EXAM TIP

If a field is conservative, do not parameterize a path unless the question specifically asks you to. Find the potential and use endpoint subtraction; it is much faster.

9. Set 8 - Surface, Volume and Line Integrals

8(a) Flux through a plane in the first octant

$$\mathbf{A} = 18z\hat{i} - 12\hat{j} + 3y\hat{k},$$

and S is the part of

$$2x + 3y + 6z = 12$$

in the first octant.

BASIC IDEA

A surface flux integral measures how much of a vector field passes through a surface. For a surface written as $z = f(x, y)$ with upward orientation, $\mathbf{n}\, dS = (-f_x, -f_y, 1)\, dx\, dy$.

Write the plane as

$$z = 2 - \frac{x}{3} - \frac{y}{2}.$$

Then

$$z_x = -\frac{1}{3}, \qquad z_y = -\frac{1}{2},$$

so for the normal in the positive direction of $(2, 3, 6)$,

$$\mathbf{n}\, dS = \left(\frac{1}{3}, \frac{1}{2}, 1\right) dx\, dy.$$

Therefore

$$\mathbf{A} \cdot (\mathbf{n}\, dS) = 18z\left(\frac{1}{3}\right) - 12\left(\frac{1}{2}\right) + 3y = 6z - 6 + 3y.$$

Substitute $z = 2 - x/3 - y/2$:

$$6z - 6 + 3y = 6 - 2x.$$

The projection on the xy-plane is

$$D: \quad x \geq 0,\ y \geq 0,\ 2x + 3y \leq 12.$$

Thus

$$0 \leq x \leq 6, \qquad 0 \leq y \leq 4 - \frac{2x}{3}.$$

Hence

$$\iint_S \mathbf{A} \cdot \mathbf{n}\, dS = \int_0^6 \int_0^{4-2x/3} (6 - 2x)\, dy\, dx.$$

Integrating,

$$= \int_0^6 (6 - 2x)\left(4 - \frac{2x}{3}\right) dx = \boxed{24}.$$

COMMON MISTAKE

If the opposite orientation is chosen, the answer is -24. Orientation controls the sign of any flux integral.

8(b) Vector triple integral

$$\mathbf{F} = 2xz\hat{i} - x\hat{j} + y^2\hat{k},$$

with the region bounded by

$$x = 0, \quad y = 0, \quad y = 6, \quad z = x^2, \quad z = 4.$$

The surfaces $z = x^2$ and $z = 4$ meet when

$$x^2 = 4 \quad \Rightarrow \quad 0 \leq x \leq 2$$

because $x = 0$ is also a boundary. Thus a natural order is

$$0 \le x \le 2, \qquad 0 \le y \le 6, \qquad x^2 \le z \le 4.$$

Integrate component by component:

$$\iiint_V \mathbf{F}\, dV = I\hat{\imath} + J\hat{\jmath} + K\hat{k}.$$

For the $\hat{\imath}$ component,

$$I = \int_0^2 \int_0^6 \int_{x^2}^4 2xz\, dz\, dy\, dx = \boxed{128}.$$

For the $\hat{\jmath}$ component,

$$J = \int_0^2 \int_0^6 \int_{x^2}^4 (-x)\, dz\, dy\, dx = \boxed{-24}.$$

For the $\hat{k}$ component,

$$K = \int_0^2 \int_0^6 \int_{x^2}^4 y^2\, dz\, dy\, dx = \boxed{384}.$$

Therefore

$$\boxed{\iiint_V \mathbf{F}\, dV = 128\hat{\imath} - 24\hat{\jmath} + 384\hat{k}}.$$

8(c) Work once around a circle

$$\mathbf{F} = (2x - y + z)\hat{\imath} + (x + y - z^2)\hat{\jmath} + (3x - 2y + 4z)\hat{k}.$$

The circle lies in the xy-plane, is centered at the origin, and has radius 3.

BASIC IDEA

On a closed curve in the xy-plane, $z = 0$ and $dz = 0$. Green's theorem converts $\oint_C P\,dx + Q\,dy$ into a double integral: $\iint_D (Q_x - P_y)dA$.

On $z = 0$,

$$P = 2x - y, \qquad Q = x + y.$$

Therefore

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 1 - (-1) = 2.$$

For counter-clockwise orientation,

$$W = \iint_D 2\, dA = 2 \times \text{area of circle}.$$

Since the radius is 3,

$$\boxed{W = 2\pi(3)^2 = 18\pi}.$$

For clockwise orientation, the answer is -18π.

10. Rapid Revision Sheet

Matrices and linear algebra

$$\begin{aligned}
\text{Idempotent: } & A^2 = A, \\
\text{Nilpotent: } & A^k = 0 \text{ for some } k, \\
\text{Symmetric part: } & \frac{1}{2}(A + A^T), \\
\text{Skew part: } & \frac{1}{2}(A - A^T), \\
A^{-1} \text{ exists: } & \det A \neq 0, \\
\text{Rank: } & \text{number of pivots}, \\
\text{Eigenvalues: } & \det(A - \lambda I) = 0.
\end{aligned}$$

Laplace transform

$$\begin{aligned}
\mathcal{L}\{\sin at\} &= \frac{a}{s^2 + a^2}, \\
\mathcal{L}\{tf(t)\} &= -F'(s), \\
\mathcal{L}\{e^{-at}f(t)\} &= F(s+a), \\
\mathcal{L}\{f(t-a)u(t-a)\} &= e^{-as}F(s), \\
\mathcal{L}\{f\}_{\text{period } T} &= \frac{\int_0^T e^{-st} f(t)dt}{1 - e^{-sT}}, \\
(f * g)(t) &= \int_0^t f(\tau)g(t-\tau)d\tau.
\end{aligned}$$

Vector calculus

$$\begin{aligned}
\nabla T &= (T_x, T_y, T_z), \\
\nabla \cdot \mathbf{V} &= P_x + Q_y + R_z, \\
\nabla \times \mathbf{V} &= (R_y - Q_z,\ P_z - R_x,\ Q_x - P_y), \\
\text{Conservative: } \mathbf{F} &= \nabla\phi \quad \Rightarrow \quad W = \phi(B) - \phi(A), \\
\text{Green: } \oint_C P dx + Q dy &= \iint_D (Q_x - P_y) dA.
\end{aligned}$$

Final answers at a glance

Question	Final result
1(b)	Idempotent: $A^2 = A$
1(c)	$S = \begin{bmatrix} 3 & 3 & 1 \\ 3 & 0 & 7/2 \\ 1 & 7/2 & 7 \end{bmatrix}, K = \begin{bmatrix} 0 & -2 & -1 \\ 2 & 0 & -1/2 \\ 1 & 1/2 & 0 \end{bmatrix}$
1(d)	$(x, y, z) = (2, 1, -1)$
2(a)	$A^{-1} = \begin{bmatrix} 11/79 & 2 & -62/79 \\ 4/79 & 0 & -1/79 \\ -6/79 & -1 & 41/79 \end{bmatrix}$
2(b)	Unique: $a \neq -3$; none: $a = -3, b \neq 1/3$; infinite: $a = -3, b = 1/3$

2(c)	Vectors are linearly independent
3(a)	$\text{rank}(A) = 4$
3(b)	Eigenvalues $8, 2, 2$; one eigenvector for 8 is $(2, -1, 1)^T$
4(a)	$\lvert v(0)\rvert = \sqrt{37}$, $\lvert a(0)\rvert = 5\sqrt{13}$
4(b)	$W = 10$
4(c)	$A^{-1} = \frac{1}{3}\begin{bmatrix} 1 & -2 \\ 1 & 1 \end{bmatrix}$
5(a)(i)	$\dfrac{10(s+3)}{((s+3)^2+25)^2} + \dfrac{10}{(s+3)^2+25}$
5(a)(ii)	$\dfrac{1-(1+2s)e^{-2s}}{s^2(1-e^{-4s})}$
5(b)	$p(t) = 5t - 5(t-4)u(t-4)$, $P(s) = 5(1-e^{-4s})/s^2$
5(c)(i)	$\dfrac{(t-5)^3}{6}e^{2(t-5)}u(t-5)$
5(c)(ii)	$-e^{-t}/6 - 4e^{2t}/3 + 7e^{3t}/2$
6(a)	$(e^{2t} - e^{-2t})/16 - (t/4)e^{-2t}$
6(b)	$x(t) = e^{-t}(t + t^2/2)$
6(c)	$\lvert\nabla T\rvert = \sqrt{429}$; direction $(16, 13, -2)/\sqrt{429}$
7(a)	Divergence $= 13 > 0$: expansion
7(b)	Irrotational; $\phi = (x^4 + y^4 + z^4)/4 + xyz + C$
7(c)	Conservative; work $= 7$
8(a)	Flux $= 24$ for positive normal; -24 for opposite normal
8(b)	$128\hat{i} - 24\hat{j} + 384\hat{k}$
8(c)	18π counter-clockwise; -18π clockwise

11. A Simple Study Order for a Beginner

If you are starting almost from zero, use this order:

1. **First session:** Set 1 and Set 2. Learn matrix multiplication, determinant, inverse, and row operations.
2. **Second session:** Set 3. Practice row reduction and the eigenvalue equation.
3. **Third session:** Set 5. Memorize the small Laplace table and both shifting rules.
4. **Fourth session:** Set 6. Learn convolution and the Laplace method for differential equations.
5. **Fifth session:** Set 7. Memorize gradient, divergence, and curl formulas and understand what each means.
6. **Sixth session:** Set 8. Practice choosing bounds before doing any integration.
7. **Final revision:** Set 4 and then solve the full paper without looking at this guide.

EXAM TIP

The fastest way to improve is to redo the same questions 24 hours later without looking at the solution. If you can reproduce the procedure, you are learning the topic; if you only recognize the answer, you are memorizing.

Prepared as a beginner-friendly study aid from the two examination-paper photographs supplied in this chat.